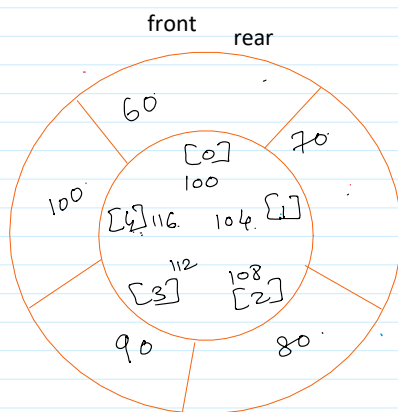


int arr[5]

9 full

$$* \text{front} = (\text{Rear} + 1) \% \text{SIZE}$$

$$1. = (0 + 1) \% 5 \rightarrow 1$$

$$2. = (1 + 1) \% 5 \rightarrow 2$$

$$3. = (2 + 1) \% 5 \rightarrow 3$$

$$4. = (3 + 1) \% 5 \rightarrow 4$$

$$0 = (4 + 1) \% 5 \rightarrow 0$$

$$0 = 0$$

Enqueue:

- 1) Increment rear as

$$\text{Rear} = (\text{rear} + 1) + \text{SIZE}$$

1. Add element at rear position
2. If front == -1, make front = 0

Dequeue :

- 1) Increment front .

If front is 4 and rear is 0
to delete the rear position, we cannot
increment front as front ++ Will be
index 5 but we want to delete index 0;

So,

$$\text{Front} = \text{front} + 1 \% \text{SIZE}$$

If deleting the last element in queue

If (front == rear)

Front = rear = -1

Queue Empty condition

If (rear == -1) queue is empty

Queue Full condition :

When Queue is full,

Rear = 0 front = 1

Rear = 1, front = 2

Rear = 2 front = 3

Rear = 3 front = 4

rear = 4, front = 0

This means,

$$\text{Front} == \text{rear} + 1$$

But when rear = 4 front = 0

0 == 4+1 does not satisfy the above condition

So the queue full condition can be

$$\text{Front} == (\text{rear} + 1) \% \text{SIZE}$$

$$0 == (4 + 1) \% 5$$

$$0 == 5 \% 5$$

$$0 == 0$$

Option 2:

If rear == SIZE-1

Rear = 0;

Else

Rear++;

Enqueue (val)

① rear++; ✓

↳

$$\text{rear} = (\text{rear} + 1) \% \text{SIZE}$$

$$= (-1 + 1) \% 5$$

$$\text{rear} = 0 \% 5$$

$$(0 + 1) \% 5$$

$$\text{rear} = 1$$

$$(1 + 1) \% 5$$

$$2 \% 5$$

$$\text{rear} = 2$$

$$(2 + 1) \% 5$$

$$3 \% 5$$

$$\text{rear} = 3$$

$$(3 + 1) \% 5$$

$$4 \% 5$$

$$\text{rear} = 4$$

$$(4 + 1) \% 5$$

$$5 \% 5$$

$$\underline{\underline{\text{rear} = 0}}$$

dequeue ()

{

✓ arr[front] = 0;

✓ if (front == rear)

{

front = -1;

rear = -1;

,

Else
Rear++;

2 front = -1;
rear = -1;

}

else.

{ front = (front + 1) % SIZE;

}

3.

(0+1) % 5
1 % 5 = 1

front = 1.

(1+1) % 5

2 % 5

front = 2.

(4+1) % 5

5 % 5

front = 0.